

Chapter 2 — Exercises

Multiple-Choice Questions

Supervised learning algorithms

How to Use These Slides

- **12 multiple-choice** questions on Chapter 2 concepts
- Pick **one** answer (A–D) for each question
- The **answer key** with brief explanations is at the end
- Try to answer before revealing the solutions

Topics: classification vs regression, overfitting, k-NN, linear models, trees & ensembles, SVMs, neural nets, and uncertainty.

Q1. Which task is a *regression* problem?

- A. Deciding whether an email is spam or not
- B. Classifying the species of an iris
- C. Predicting the sale price of a house
- D. Detecting whether a transaction is fraud

Q2. "Overfitting" means a model is:

- A. too complex and memorizes training noise, generalizing poorly
- B. too simple to capture the real pattern
- C. always bad on the training data
- D. built with too few features

Q3. With k-NN, a very small k (e.g. 1) tends to give:

- A. a very smooth, simple decision boundary
- B. a complex, jagged boundary that can overfit
- C. identical predictions everywhere
- D. a strictly linear boundary

Q4. In Ridge regression, increasing α :

- A. removes the intercept term
- B. always improves the test score
- C. makes coefficients larger
- D. shrinks coefficients toward zero (a simpler model)

Q5. A distinctive property of Lasso (L1) is that it:

- A. never changes the coefficients
- B. sets some coefficients exactly to zero (feature selection)
- C. requires the target to be scaled
- D. only works for classification

Q6. Despite its name, LogisticRegression is:

- A. a clustering method
- B. a regression model for continuous output
- C. a classification algorithm
- D. a preprocessing transformer

Q7. For LogisticRegression / LinearSVC, a high value of C:

- A. fits the training data harder (less regularization)
- B. forces all coefficients to zero
- C. has no effect on the model
- D. always causes underfitting

Q8. Which models are sensitive to feature scaling?

- A. Decision trees
- B. Random forests
- C. Gradient boosting
- D. SVMs, k-NN, and neural networks

Q9. An unrestricted decision tree on the training data typically:

- A. underfits badly
- B. reaches 100% training accuracy and overfits
- C. cannot be constructed
- D. ignores the target values

Q10. Random forests reduce overfitting mainly by:

- A. using a single very deep tree
- B. removing all features
- C. averaging many trees trained on random subsets
- D. scaling the data first

Q11. Gradient boosted trees are built:

- A. serially, each tree correcting the previous ones' errors
- B. all at once in parallel, then averaged
- C. with no depth limit by default
- D. only for regression tasks

Q12. `predict_proba` returns:

- A. the single predicted class label
- B. a probability per class, summing to 1
- C. the distance to the decision boundary
- D. the number of support vectors

Answer Key

Solutions & Explanations

Answers 1–6

Q	Ans	Why
1	C	House price is a continuous value → regression
2	A	Overfitting = too complex, memorizes noise, poor generalization
3	B	Small <code>k</code> → complex, jagged boundary (high variance)
4	D	Higher <code>alpha</code> → more L2 regularization → smaller coefficients
5	B	L1 drives some coefficients to zero → feature selection
6	C	<code>LogisticRegression</code> is a classifier , not a regressor

Answers 7–12

Q	Ans	Why
7	A	High <code>c</code> → less regularization → fits training data harder
8	D	SVMs, k-NN, and neural nets need scaled features
9	B	A full tree splits until pure → 100% train, overfits
10	C	Averaging many randomized trees cancels overfitting
11	A	Boosting builds trees serially , fixing prior errors
12	B	<code>predict_proba</code> → per-class probabilities summing to 1

Well done!

Review Chapter 2 for any you missed